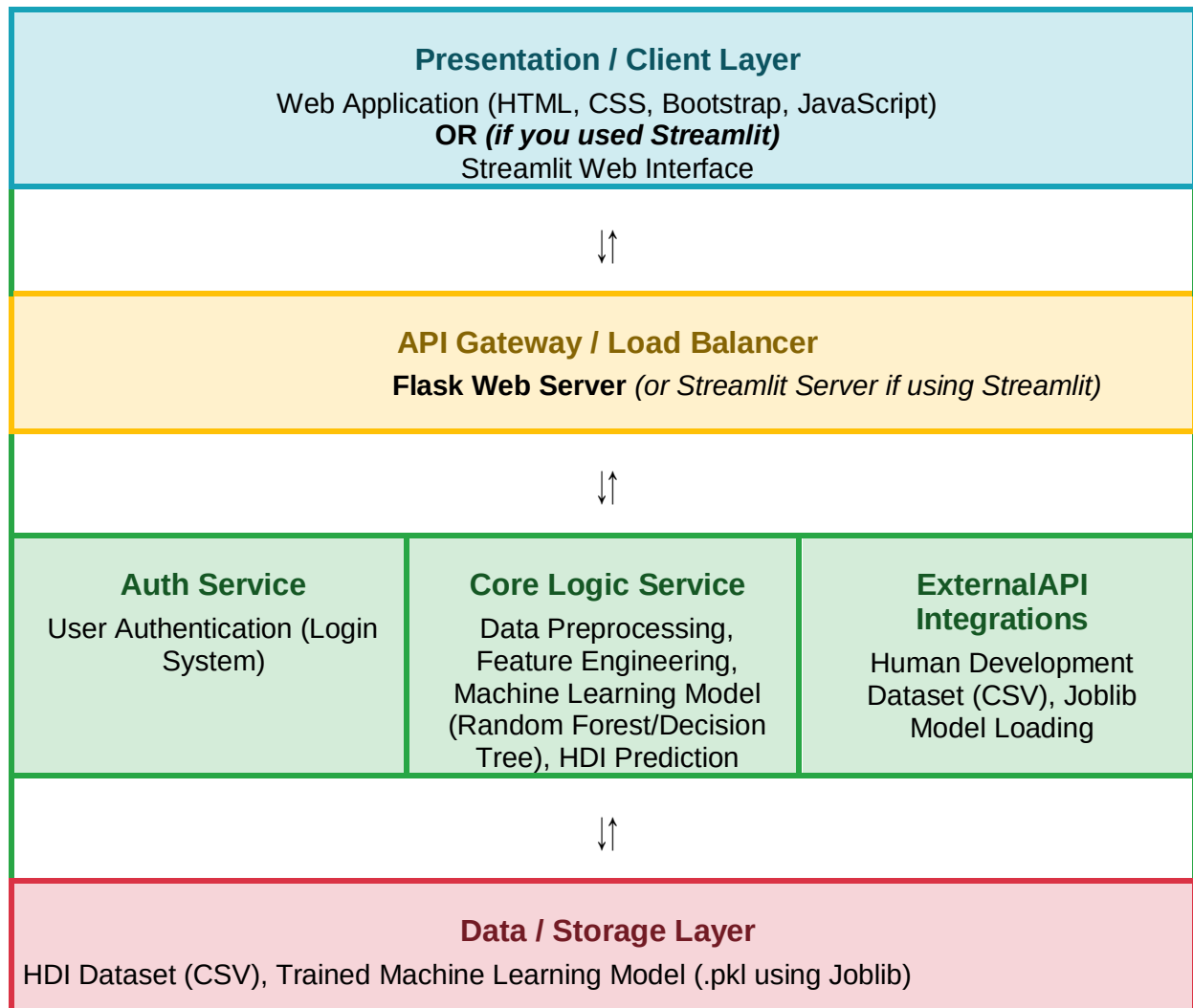


Solution Architecture

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description/Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface where users enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita to receive HDI predictions.	HTML, CSS, Bootstrap, JavaScript (or Streamlit)
API Gateway	Receives user requests, validates input data, communicates with the machine learning model, and returns prediction results to the user.	Flask (or Streamlit Server)
Micro services	Performs data preprocessing, loads the trained machine learning model, processes user input, predicts the HDI category, and displays the result.	Python, Scikit-learn, Pandas, NumPy, Joblib
Database	Stores the Human Development Index dataset and the trained machine learning model used for prediction.	CSV Dataset, Joblib (.pkl Model)